Universidade do Minho
Departamento de Informática

Dados e Aprendizagem Automática
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Practical Exercise no. 4

Theme Linear and Logistic Regression

Exercise Linear and logistic regression are two supervised learning techniques used in machine learning to estimate the value or class of a case study given a set of characteristics and statistical patterns analysed in past case studies. Linear regression aims to estimate a certain numerical value given a set of variables (a regression algorithm). Logistic regression, on the other hand, focuses on classifying a given case study (a classification algorithm).

Tasks

I – Linear Regression with Ecommerce Customers Dataset

An online clothing retailer intends to invest in improving one of its online sales platforms based on the income each provides. The available platforms are (1) a mobile application and (2) a web platform. A linear regression model was proposed to estimate the performance of each option and evaluate the best decision. To this end, the company provided a dataset containing the sales history of its customers and their respective information (e.g. email, address, time spent on each platform, total revenue acquired, etc.). This dataset is available at https://bit.ly/3mGDpu0.

After downloading the dataset, the intention is to:

T1. Load the dataset using the *pandas.read_csv(...)* function.

T2. Apply methods for data exploration and visualisation.

T3. Define the set of input and output variables of the model (i.e., *input = ["Avg. Session Length", "Time on App", "Time on Website", "Length of Membership"]*; *output = ["Yearly Amount Spent"]*).

T4. Prepare and organise the set of case studies from the dataset into training and test data using the *sklearn.model_selection.train_test_split(...)* function with *test_size = 0.3*.

T5. Train the linear regression model (*sklearn.linear_model.LinearRegression*) using the training data.

T6. Analyse the converged coefficients of the linear regression model and identify their significance in the context of the problem at hand.

T7. Evaluate the *Mean Absolute Error, Mean Squared Error* and *Root Mean Squared Error* of the model developed for the *'Yearly Amount Spent'* prediction (use the functions available in the *sklearn.metrics* library) and carry out the respective critical analysis.

II – Logistic Regression with Advertising Dataset

The aim of this exercise is to estimate whether a particular internet user has clicked on an advert using a logistic regression classification model. A dataset showing the habits of various internet users and illustrating a set of characteristics about each user and their decision-making is provided to develop this model. The dataset is available at https://bit.ly/3CMO63B.

After downloading the dataset, the intention is to:

T1. Load the dataset using the *pandas.read_csv(...)* function.

T2. Apply methods for data exploration and visualisation.

T3. Define the set of input and output variables of the model (i.e., *input = ["Daily Time Spent on Site", "Age", "Area Income", "Daily Internet Usage", "Male"]*; *output = ["Clicked on Ad"]*).

T4. Prepare and organise the set of case studies from the dataset into training and test data using the *sklearn.model_selection.train_test_split(...)* function with *test_size = 0.3*.;

T5. Train various logistic regression models (*sklearn.linear_model.LogisticRegression*) using the training data. Train with three different classifiers (*solver=...*).

T6. Evaluate the classification performance of the models by creating confusion matrices (*sklearn.metrics.confusion_matrix(...)*) and classification reports (*sklearn.metrics.classification_report(...)*).

T7. Given the results observed in **T6**, what conclusions can be drawn? In which situations does the model succeed or fail? How can the proposed learning model be improved? Which model (set of parameters) is the best?